Jefferson
Philadelphia University + Thomas Jefferson University

Team-Based Learning: Pathology of the Labia, Cervix & Vagina

Joanna Chan, MD

Joanna.chan@jefferson.edu
Gynecologic Pathology, Jefferson University Hospital
Associate Professor
Pathology Thread Director, SKMC Phase 2

Learning Objectives

- Describe the role of HPV in the pathogenesis of gynecologic tumors.
- Describe and identify the histopathologic features of HPV-related cervical dysplasia.
- Explain and describe the relationship between inflammatory processes of the vulva and malignancy.
- Describe and identify infectious/inflammatory processes of the vulva and cervix.

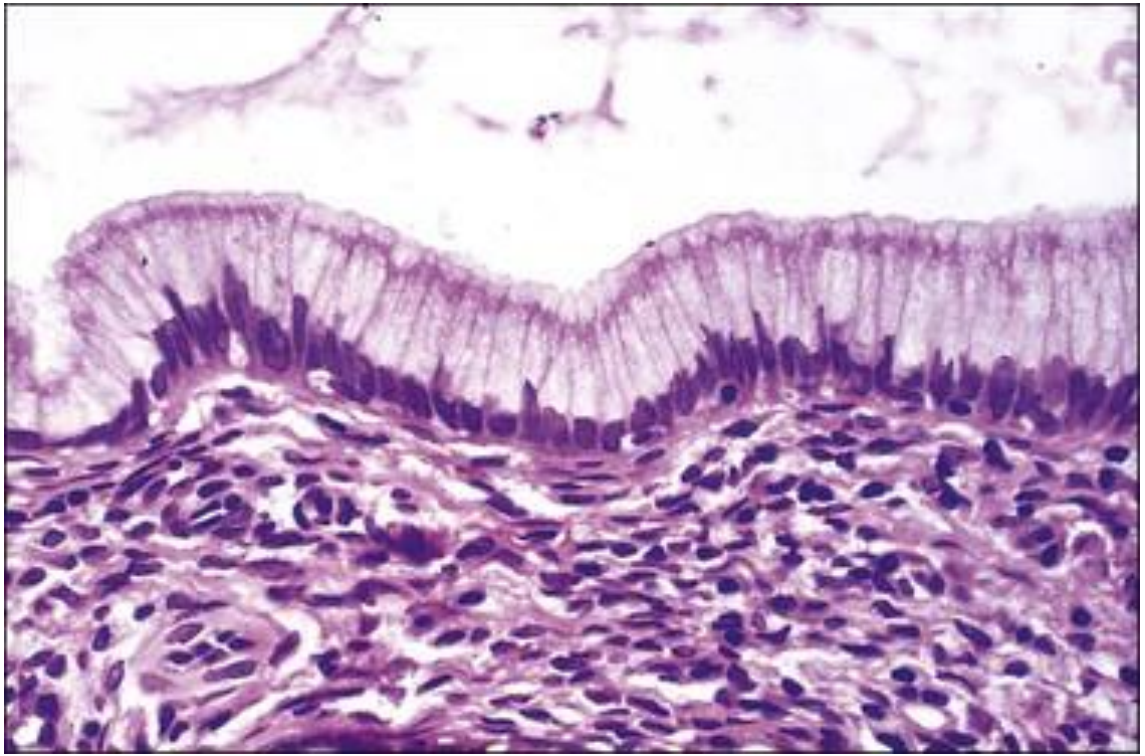
Review of Normal

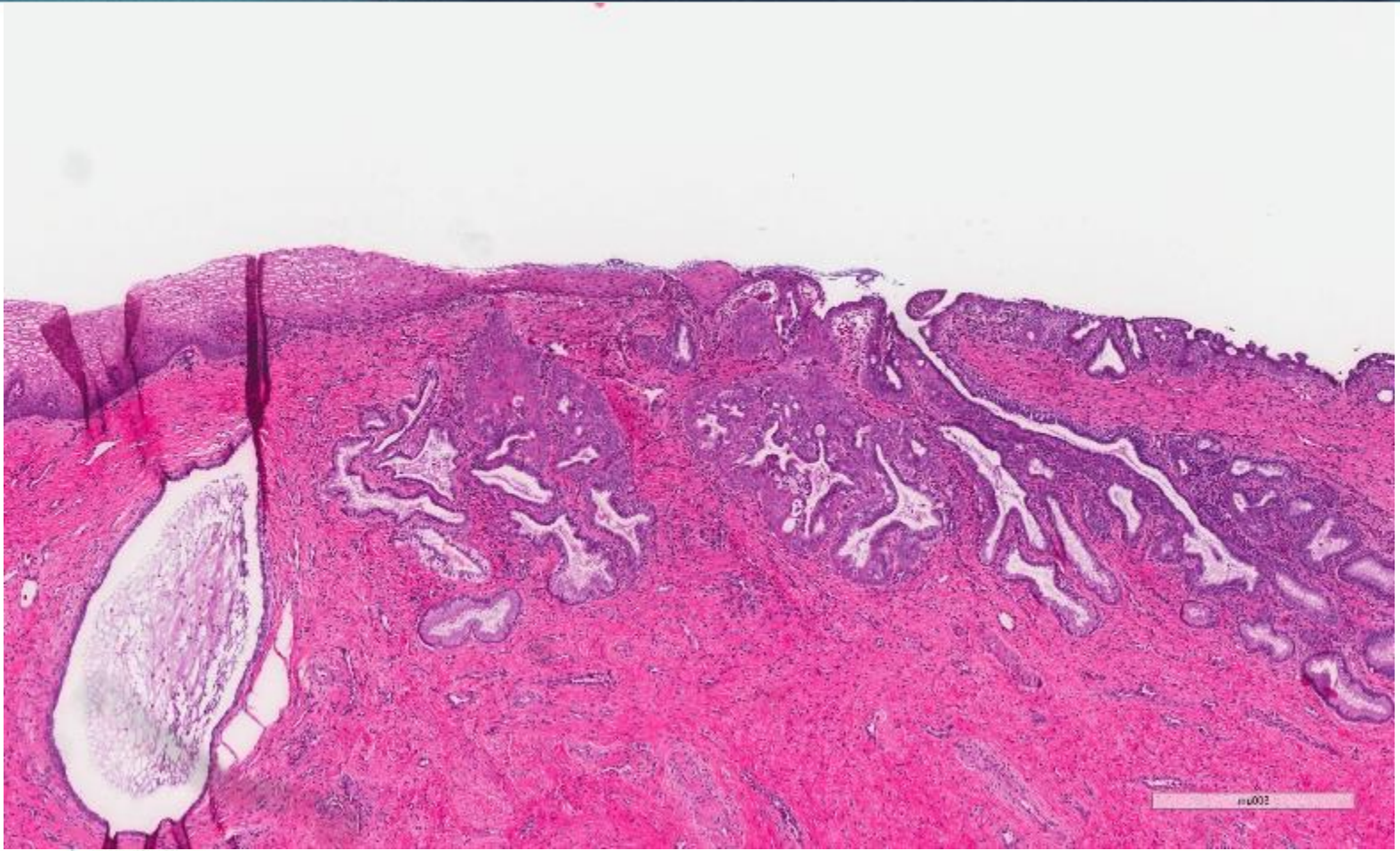


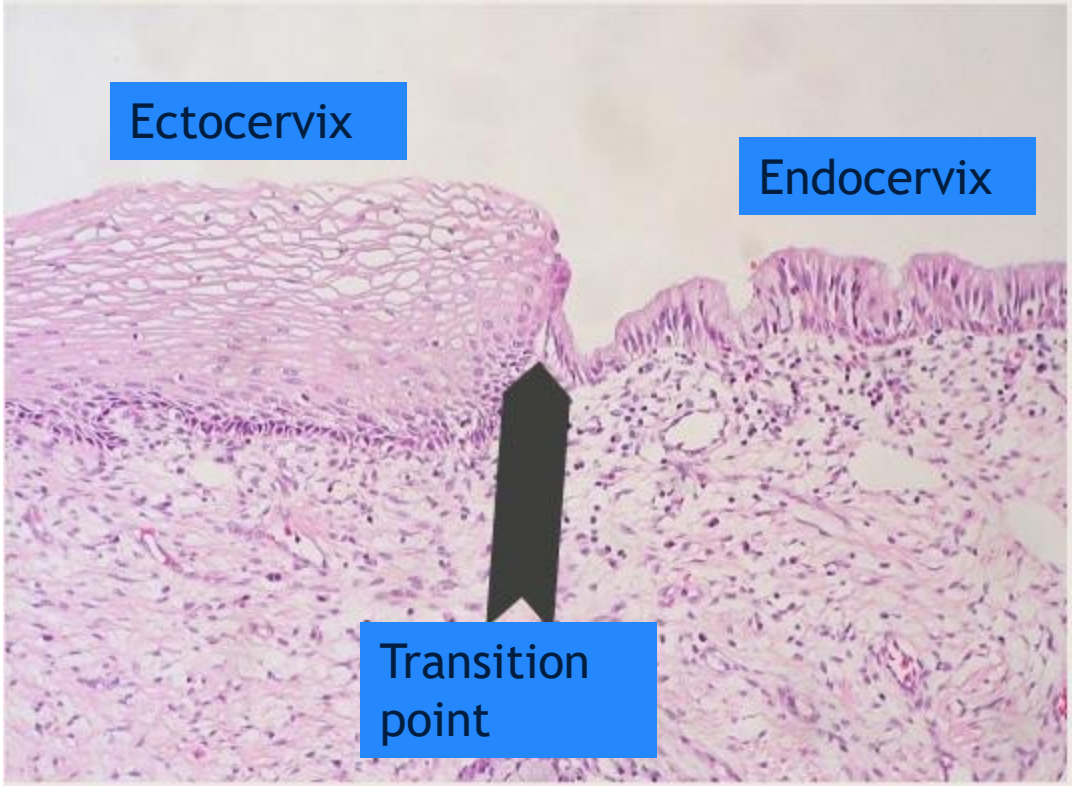


Source: Heaton RP, Smith PA, Monsour FJ. Chorney MS. The Color Atlas of Family Medicine, Second Edition: www.accessmedicine.com
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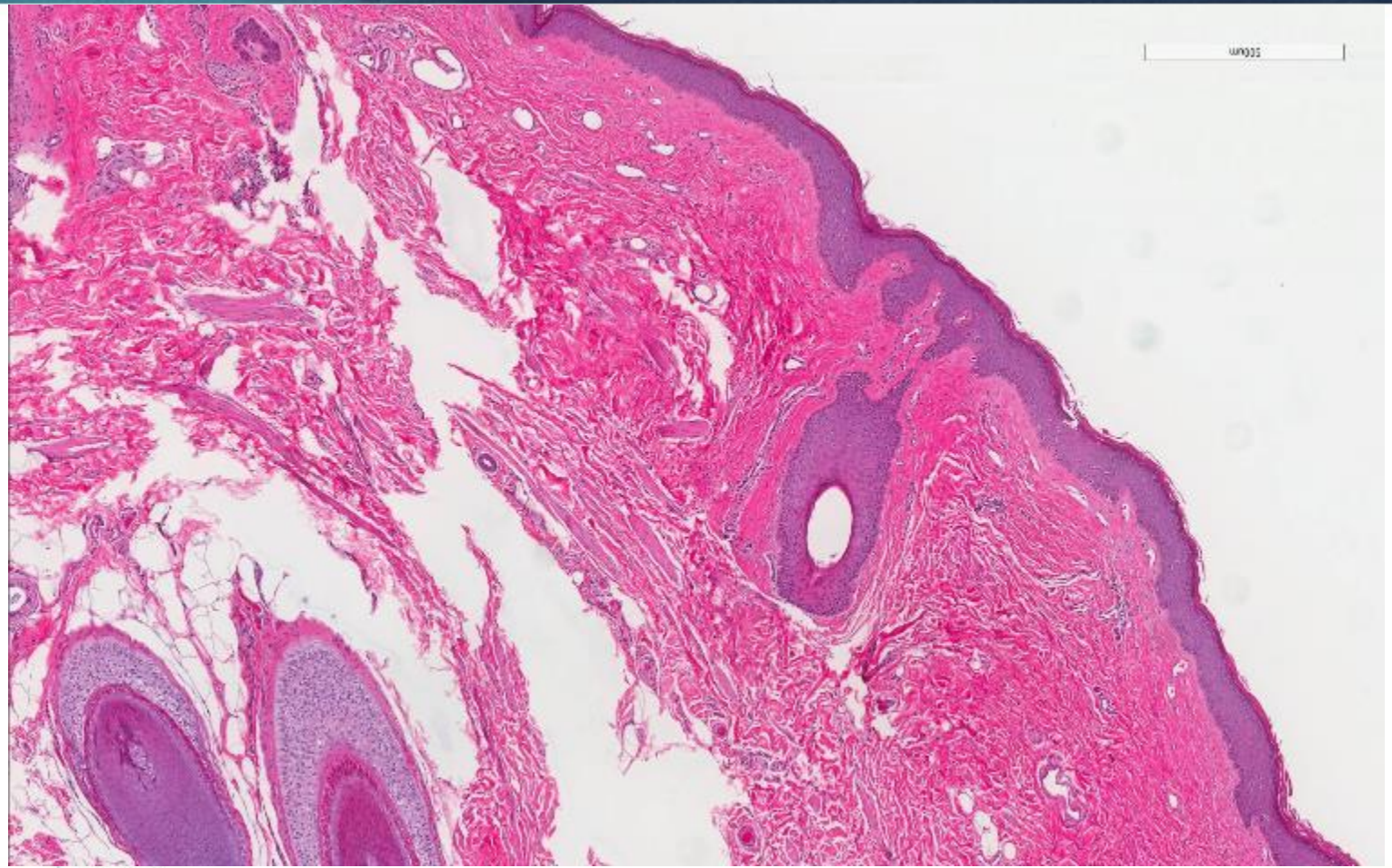





Ectocervix
Endocervix
Transition point



Normal Vulva



Case 1

Case 1

- A 34-year-old patient presents to their gynecologist for a colposcopic exam after having an “ASCUS” Pap test result. Their HPV cotesting is positive for high-risk HPV.

Case 1 Colposcopy

- ASCUS = Atypical squamous cells of uncertain significance

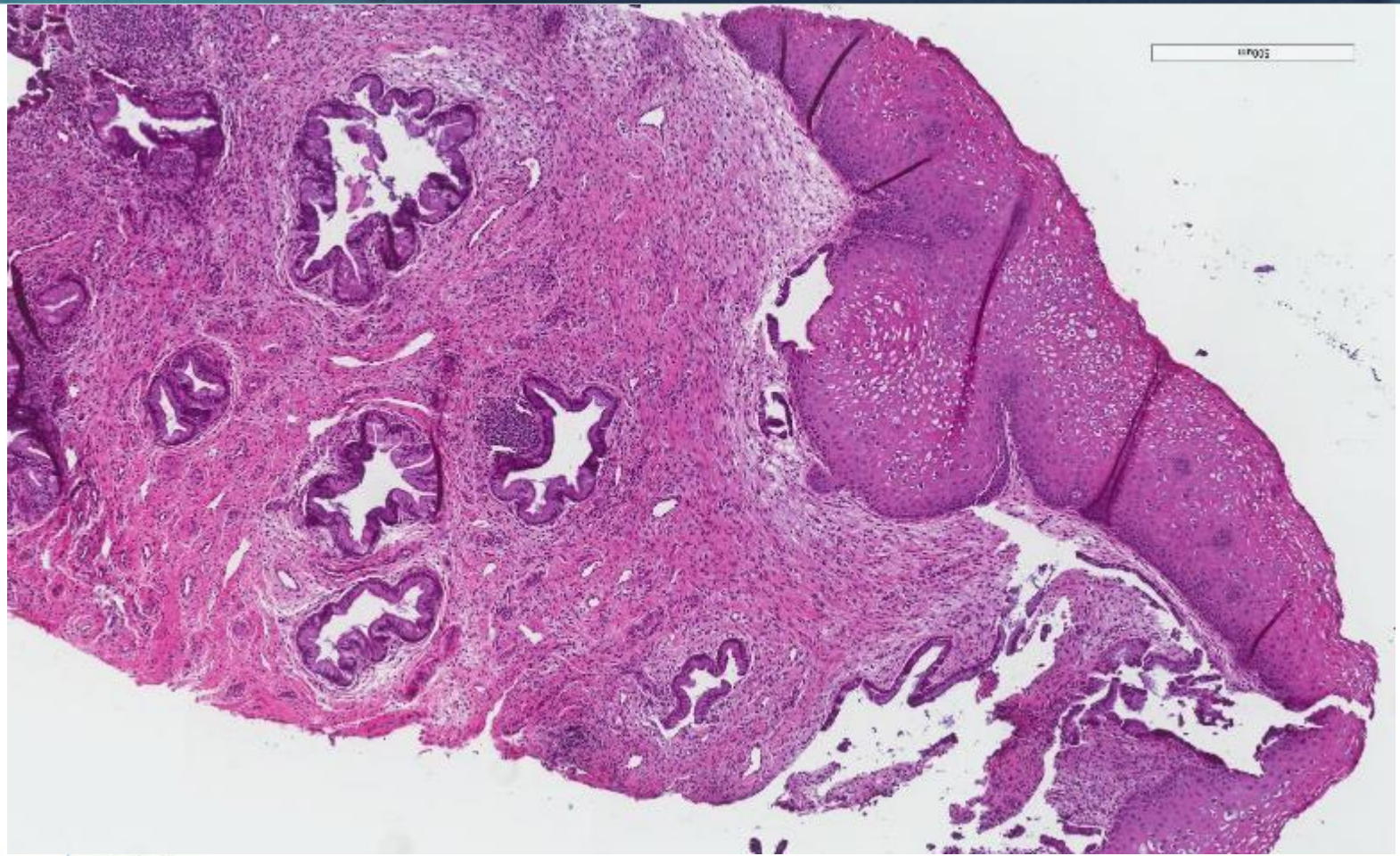


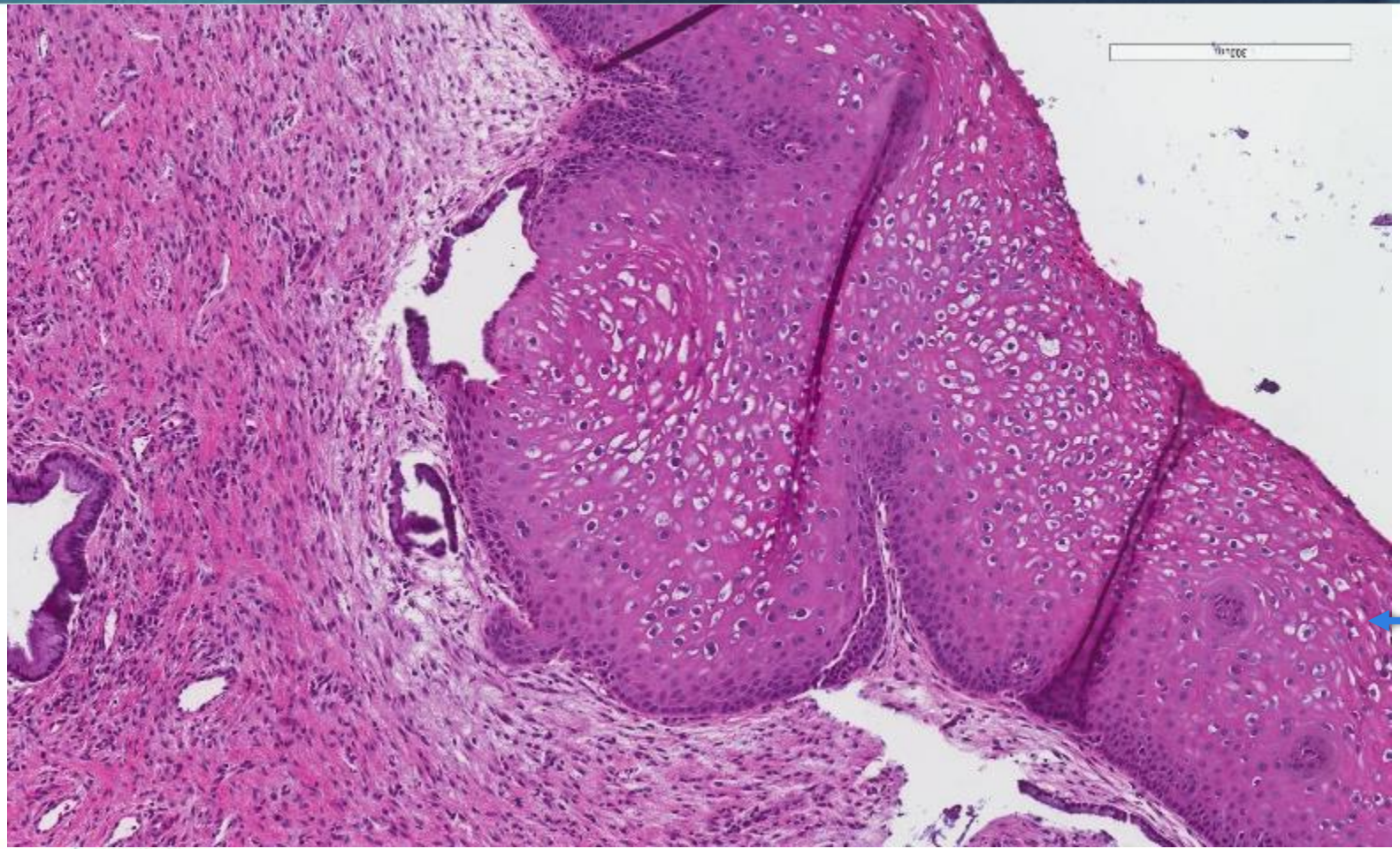
Histology time!



- Review the biopsy taken from the abnormal areas of the cervix (slide 1) and compare to normal cervix (slide 2), taking special note of:
  - Nuclear atypia of superficial squamous cells
  - Squamous maturation

Low-Grade Squamous Intraepithelial Lesion/LSIL (Slide 1)

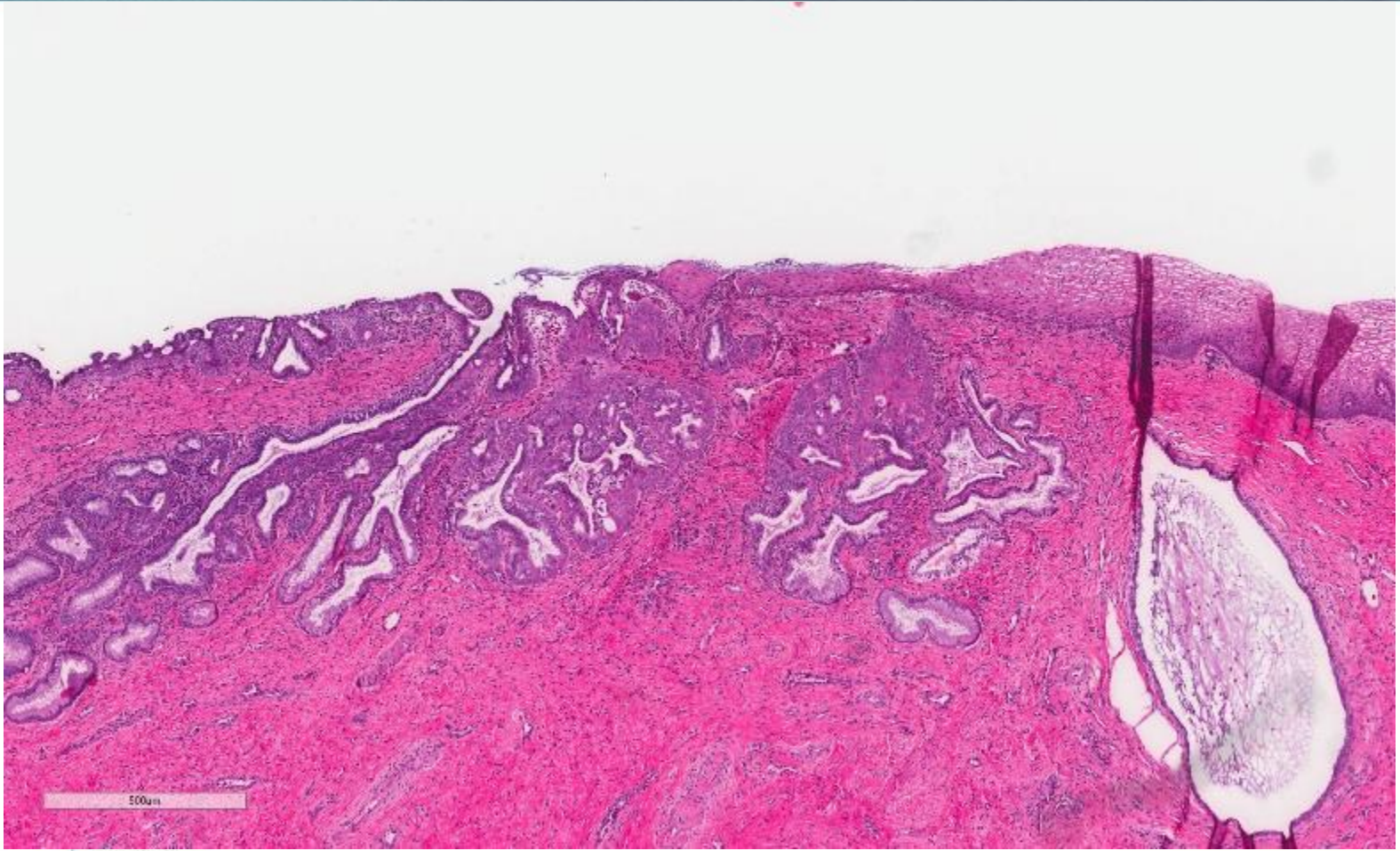



LSIL

Histology time!



- Review the biopsy taken from the abnormal areas of the cervix (slide 1) and compare to normal cervix (slide 2), taking special note of :
  - Nuclear atypia of superficial squamous cells
    - Big cells with crinkly (raisinoid) nuclei and perinuclear halos near the top
  - Squamous maturation
    - High N:C ratio (basal type cells) stay near the base, turns into lower N:C (more cytoplasm) near the surface

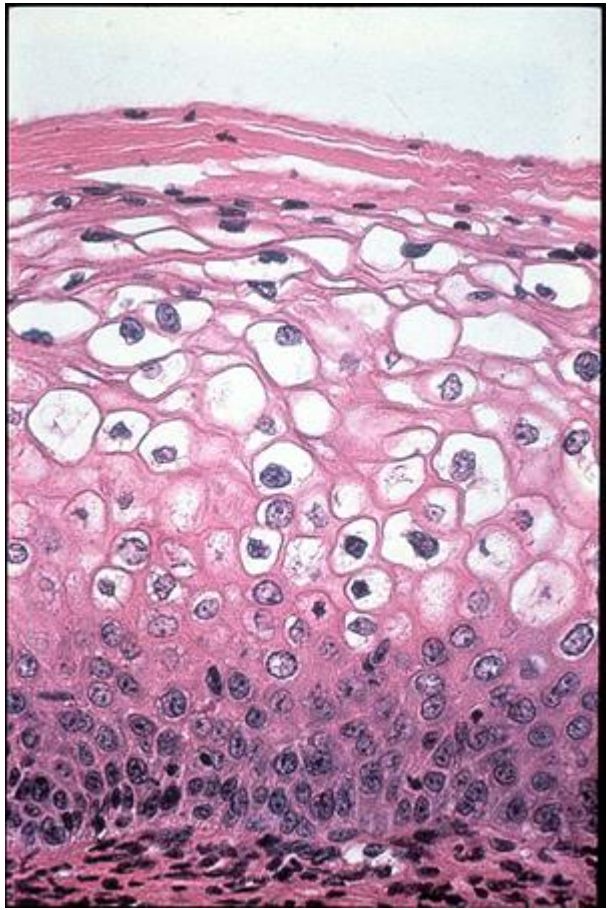

500µm

Patient Cervical Biopsy #1 vs Normal Cervix

	Patient	Normal
N:C ratio	Low	Low
Basal maturation defect	+	-
Nuclear atypia	“Koilocytic”	None
Disruption in the basement membrane	-	-
Relationship between atypical epithelium and underlying stroma	None	None

Koilocytes

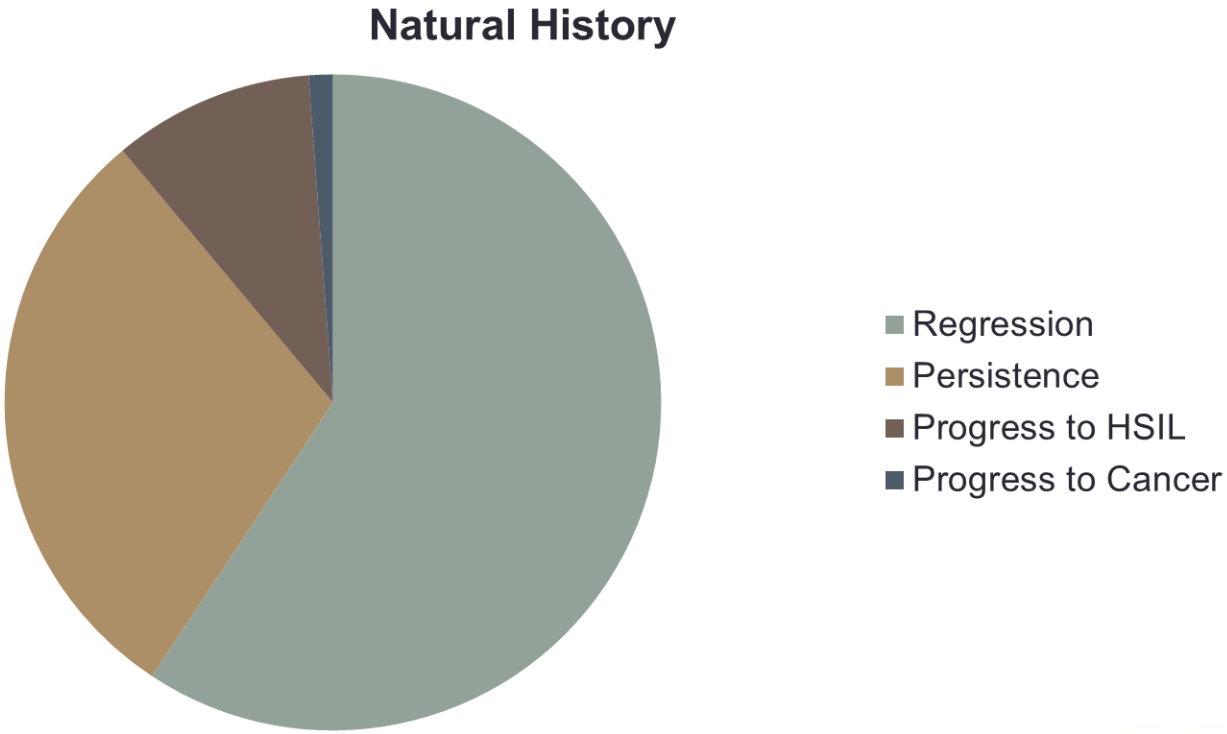
- Atypical squamous cells
- Enlarged nuclei
- Hyperchromatic
- Perinuclear halo/clearing



Diagnosis?

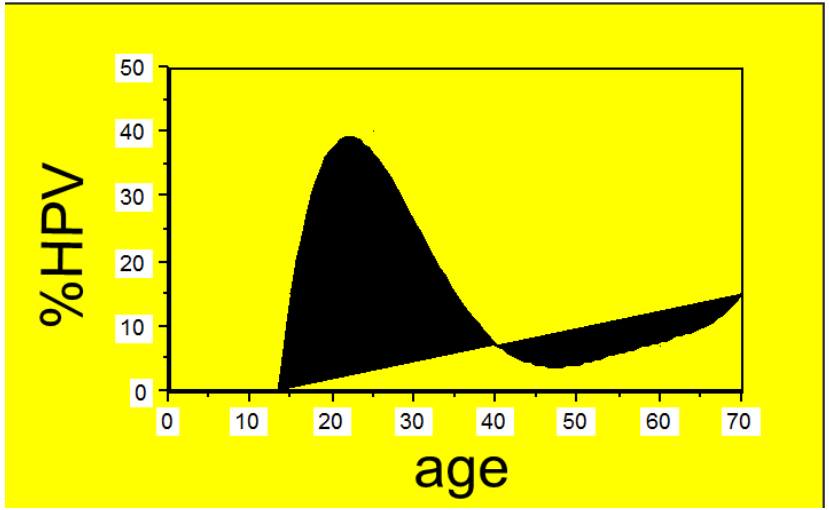
- Low-grade squamous intraepithelial lesion (LSIL)

Natural history of LSIL



How is LSIL associated with HPV?

- 100% of LSIL is associated with HPV



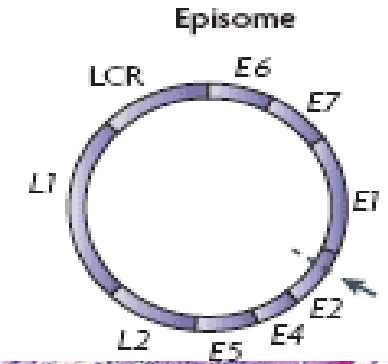
Prevalence of HPV in **normal** Pap smears by age

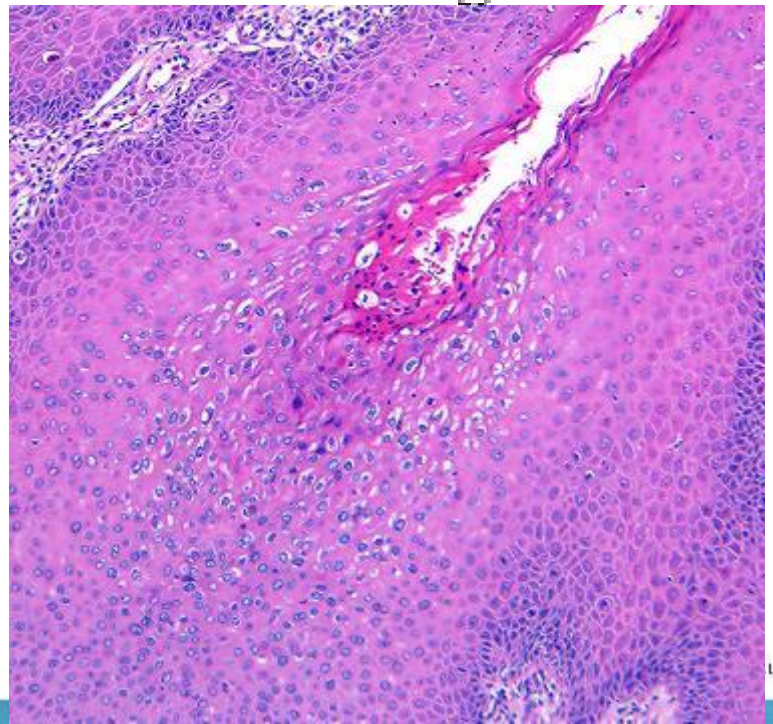
HPV Targets

- Infects immature squamous/glandular cells from the squamocolumnar junction
- Basal cells in areas of epithelial breaks

HPV in LSIL

- Viral genome remains in episomal form
- HPV gene expression only allowed in cells with squamous maturation
- Entire viral genome express → virion replication

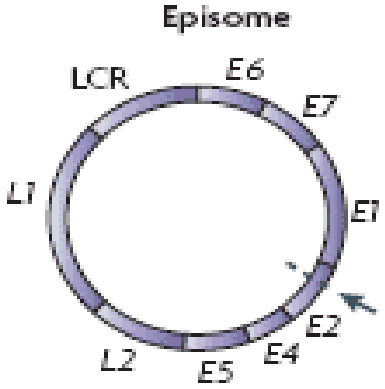


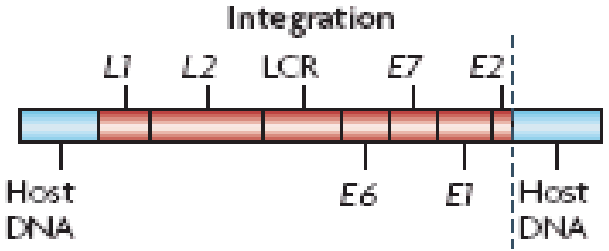


Replication of HPV results in "koilocytic atypia":
•nuclear atypia = polyploidy, aneuploidy caused by E6 & E7
•perinuclear halo = cytoskeleton disruption caused by E4

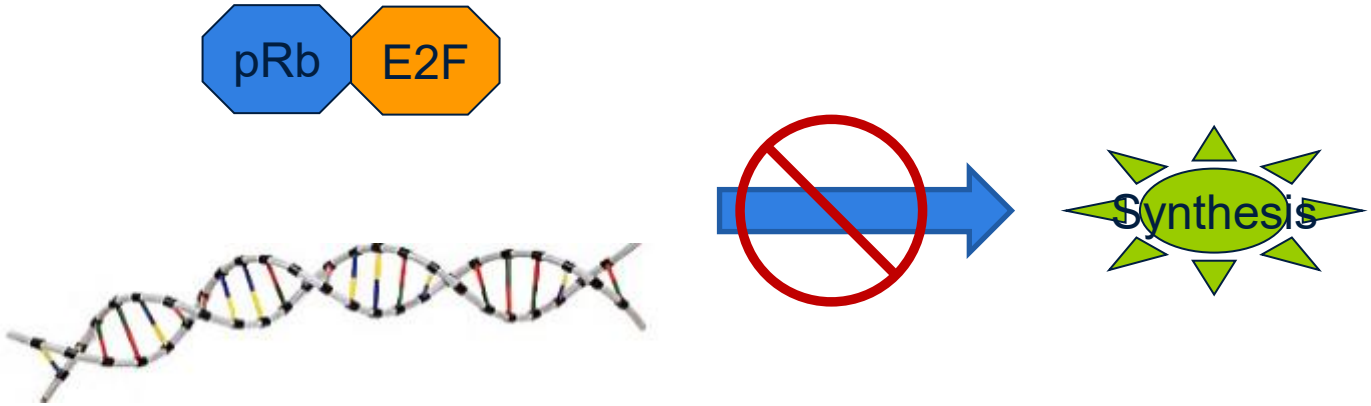
HPV in HSIL

- Virus exists in two forms in host cells
  - Episomal
  - Integrated
- Viral genome integrated into host DNA
- HPV E2 ORF disruption→ ↑ HPV oncoprotein E7
- ↑ HPV oncoprotein E7 → Uncontrolled cell proliferation

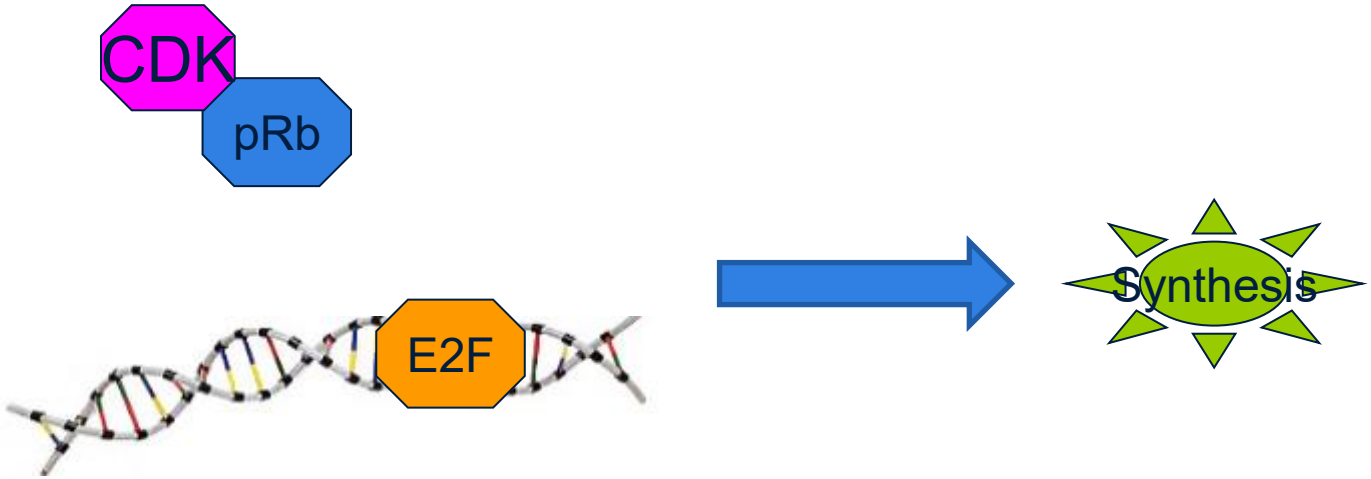




Cell Phase G_0 = NO Synthesis

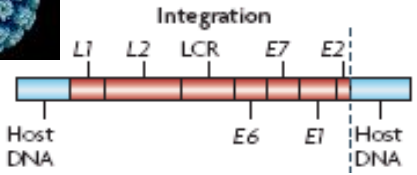


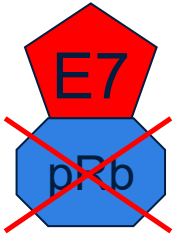
Cell Phase S = Synthesis



HPV Infection = Synthesis







HPV E2 ORF disruption→ ↑ HPV oncoprotein E7



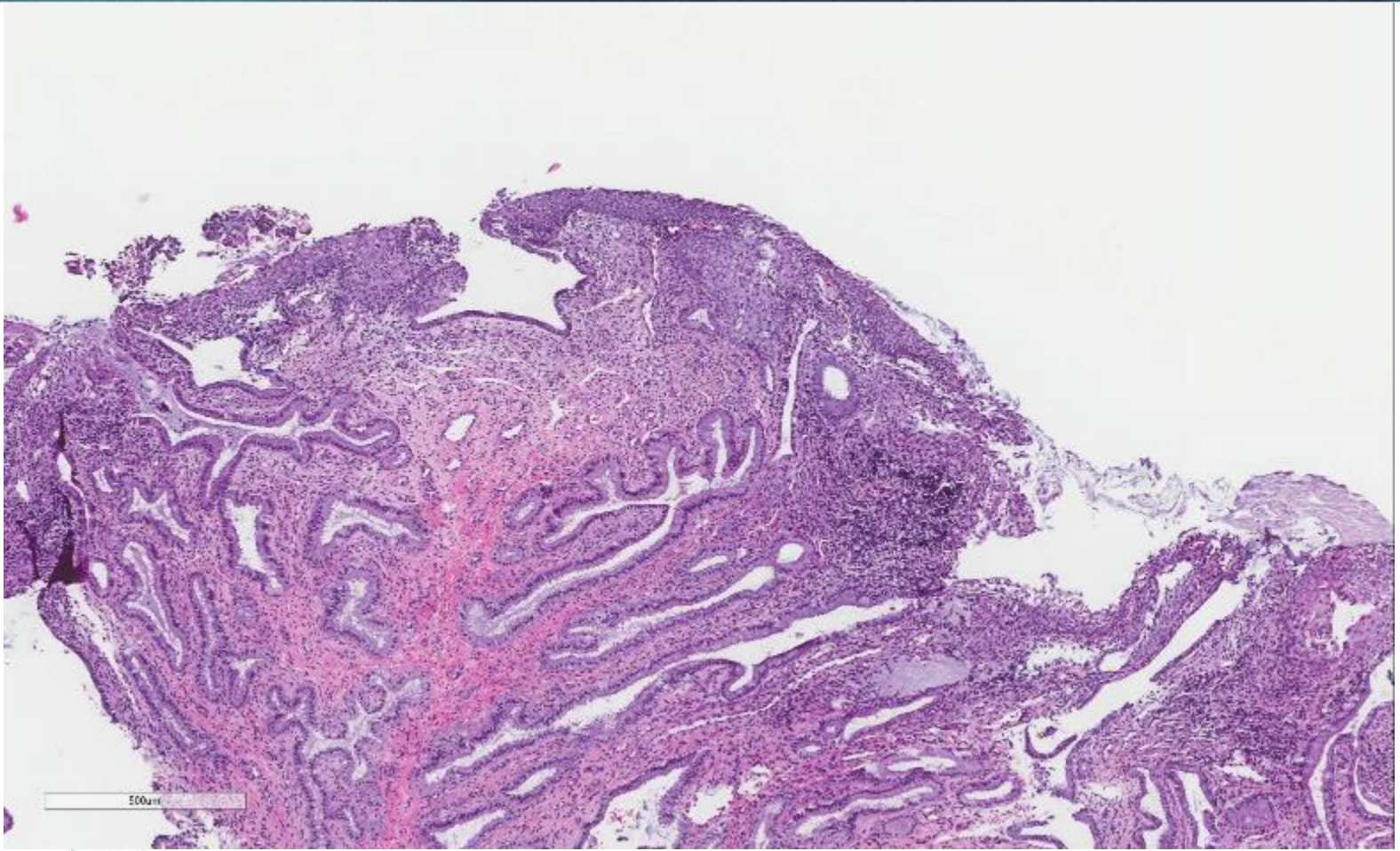


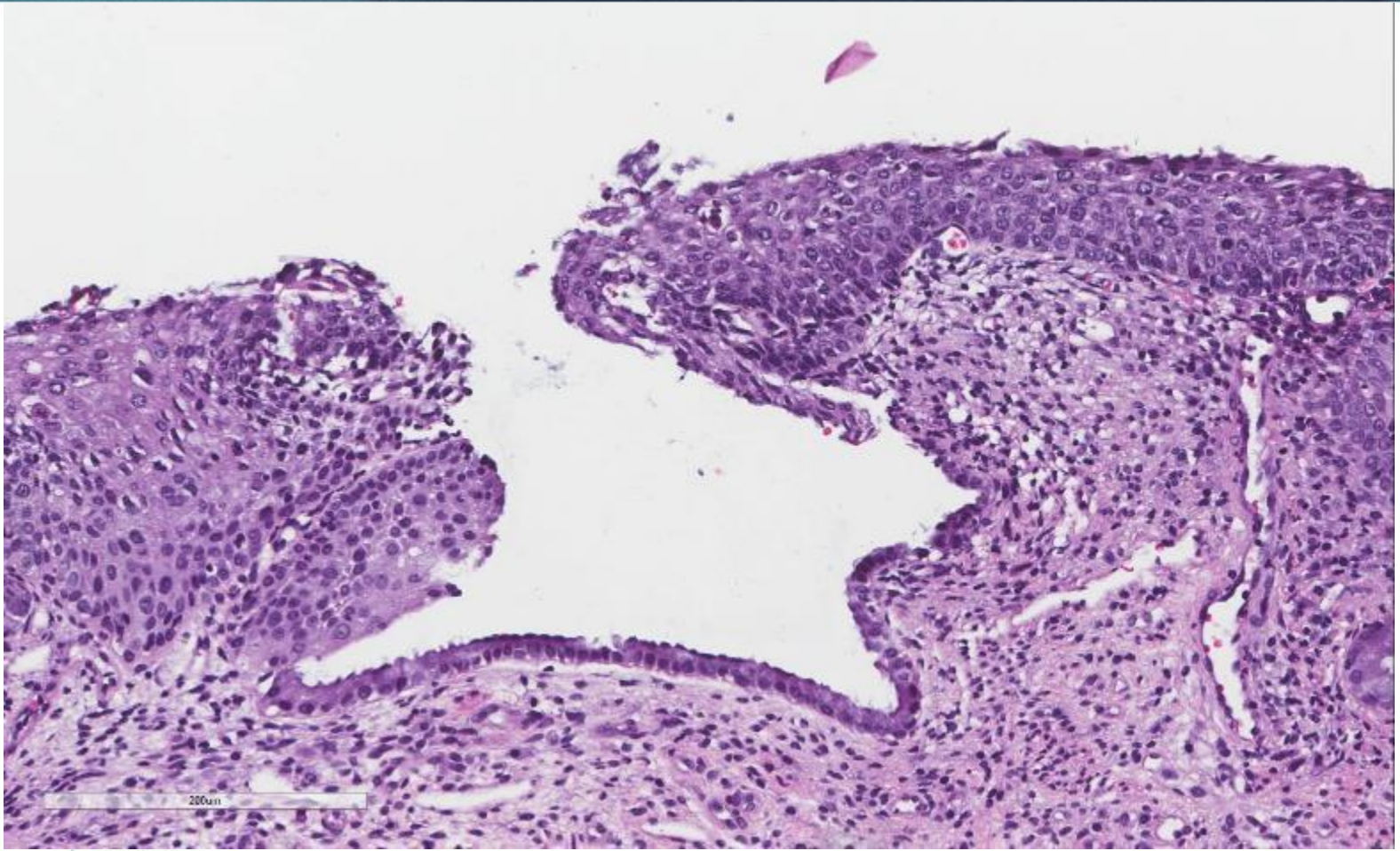


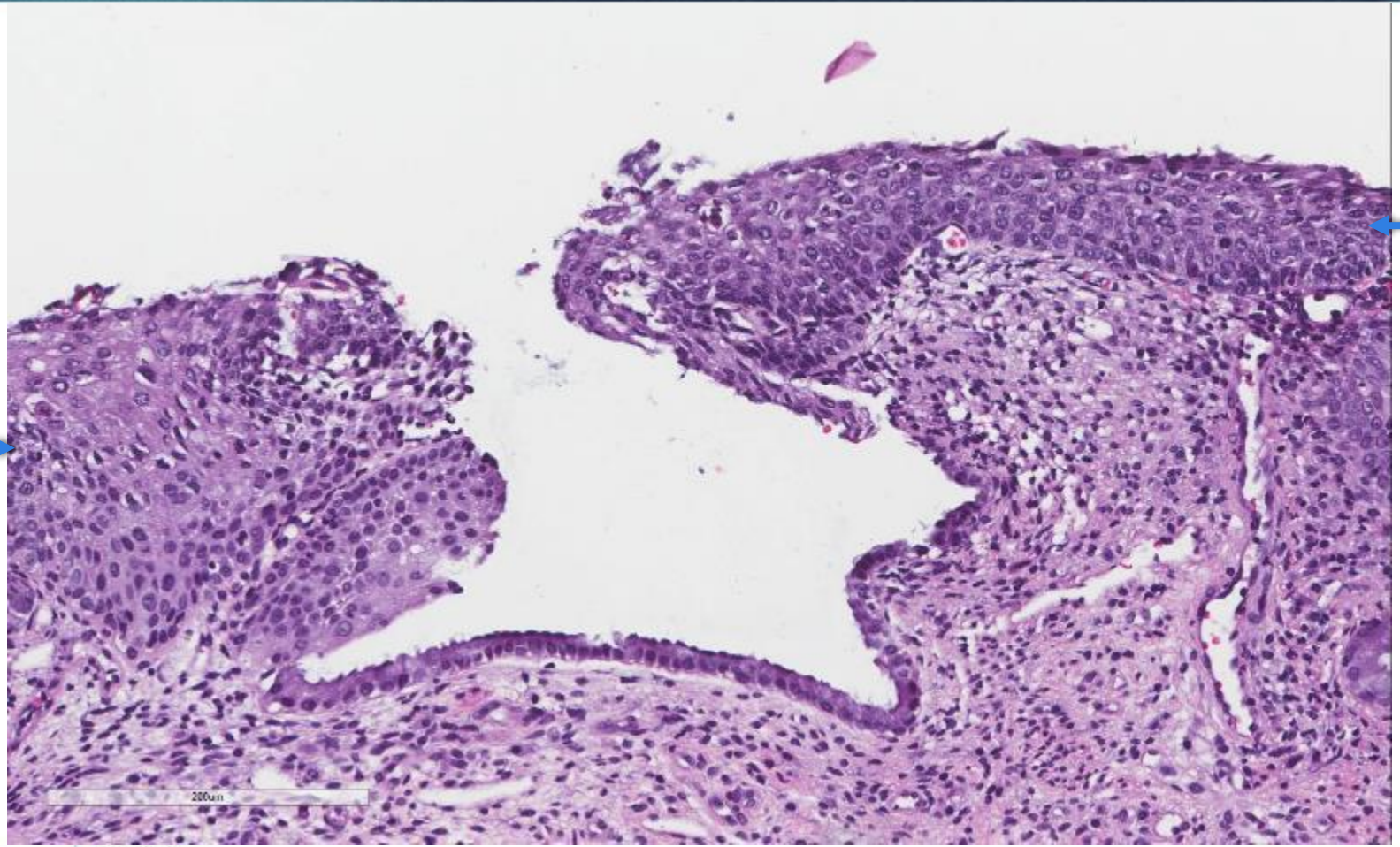
Case 1: Continued (Part 2)



- The patient is lost to follow-up for 4 years. They present again with an “atypical” Pap test and undergo a second colposcopy with associated biopsies (slide 3).
- Review their biopsy, and take particular note of:
  - Nucleus:cytoplasm (N:C) ratio
  - Basal maturation defect
  - Nuclear atypia


500µm


200um


HSIL
HSIL
200um

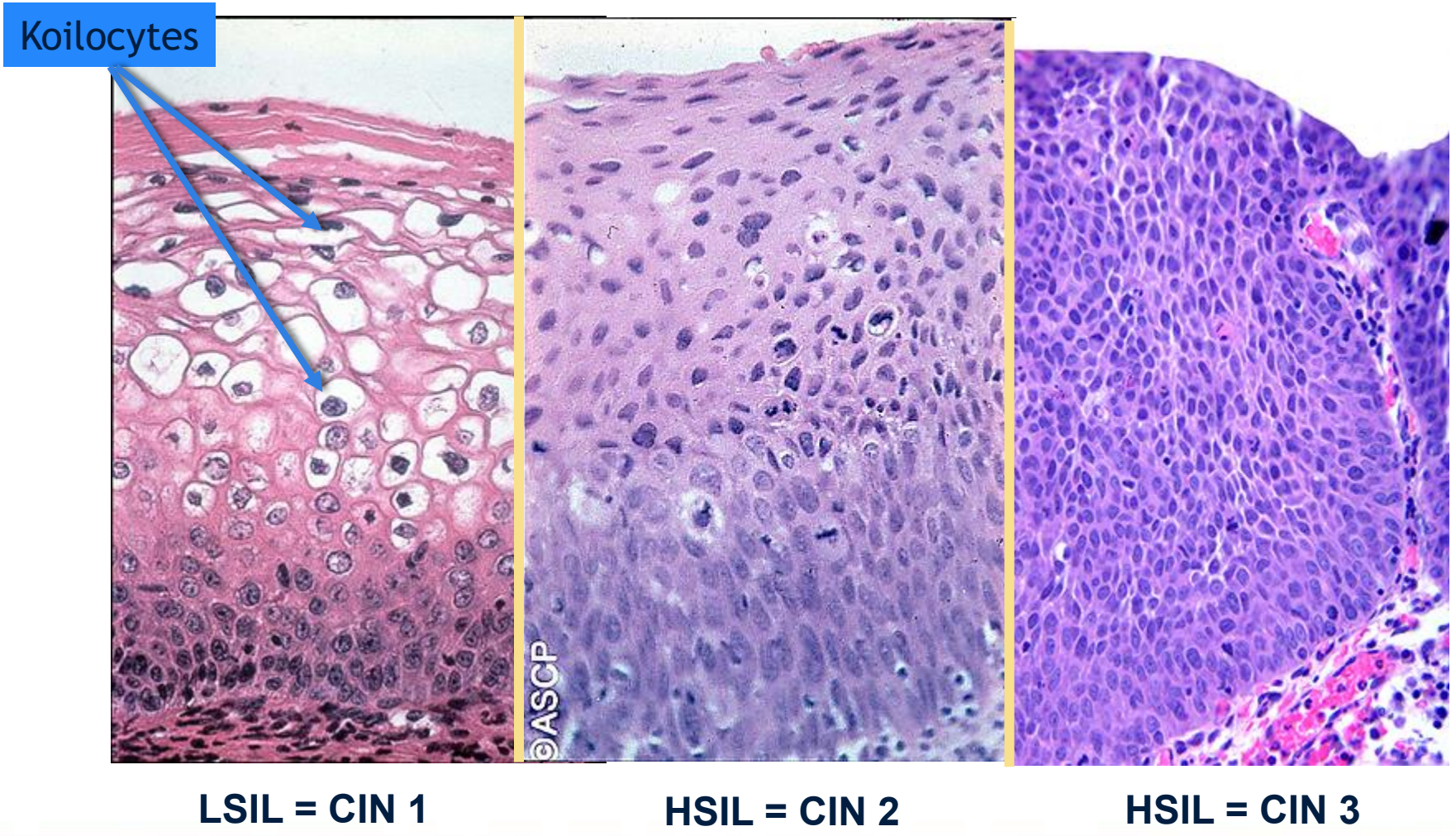
Case 1: Continued (Part 2)



- The patient is lost to follow-up for 4 years. They present again with an “atypical” Pap test and undergo a second colposcopy with associated biopsies (slide 3).
- Review their biopsy, and take particular note of:
  - Nucleus:cytoplasm (N:C) ratio
    - High N:C ratio throughout epithelium
  - Basal maturation defect
    - No maturation throughout epithelial thickness, the whole thing looks basal cells
  - Nuclear atypia
    - Not Koilocytes at the top, but not normal squamous cells

Patient Cervical Biopsy #2

	Biopsy #2	Biopsy #1	Normal
N:C ratio	High	Low	Low
Basal maturation defect	+++	+	-
Nuclear atypia	High-grade	“Koilocytic”	None
Disruption in the basement membrane	-	-	-
Relationship between atypical epithelium and underlying stroma	None	None	None



LSIL = CIN 1

HSIL = CIN 2

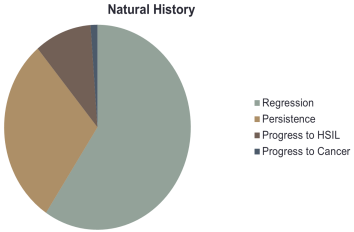
HSIL = CIN 3

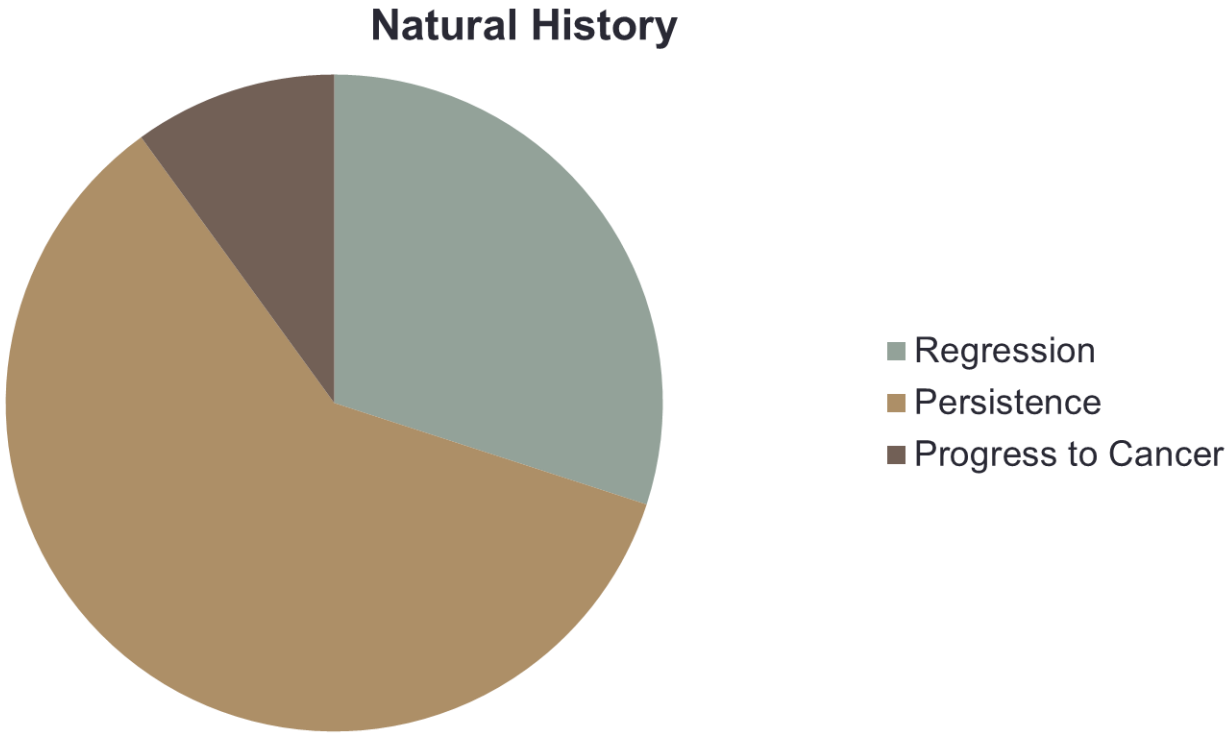
Diagnosis?

- High-grade squamous intraepithelial lesion (HSIL)

Natural history of HSIL

Natural history of LSIL

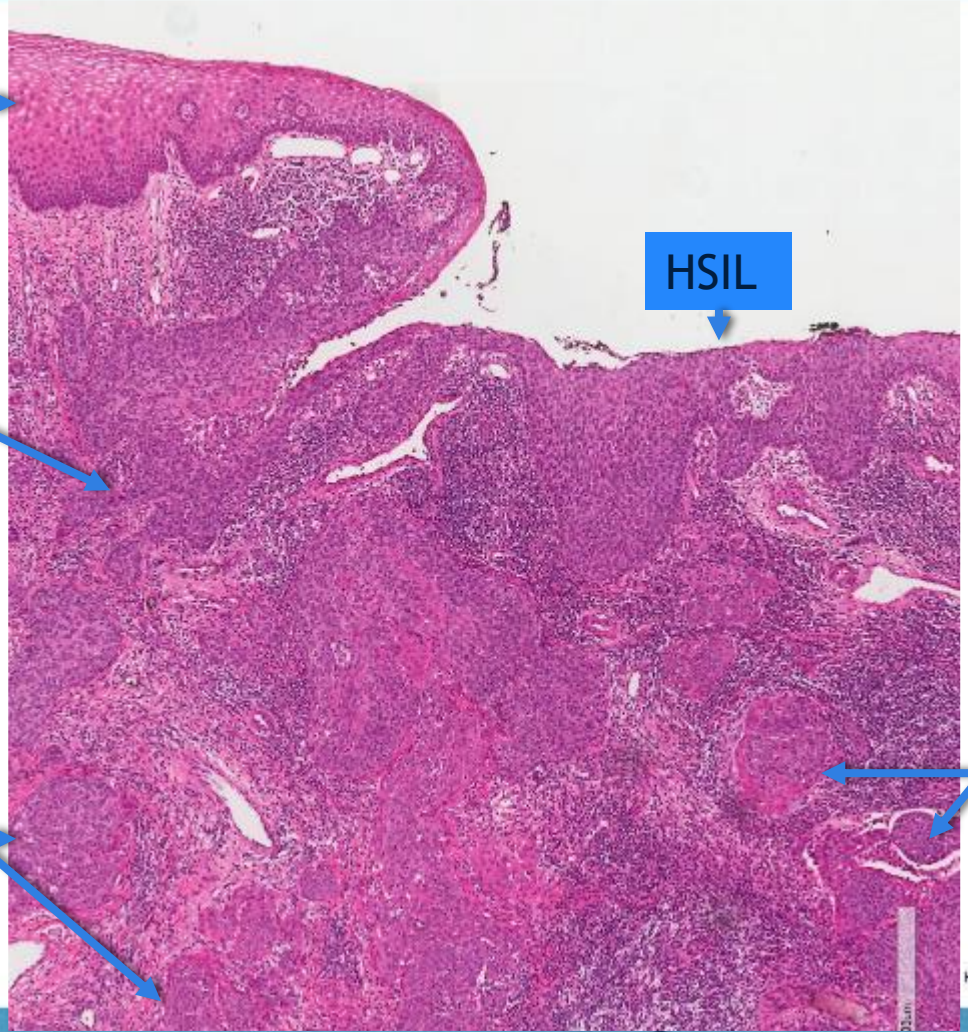




Case 1: Continued (Part 3)



- The patient is lost to follow-up for a second time. They return ten years later with vaginal bleeding. On clinical exam, they have an ulcerated cervical mass with extension into the parametrial soft tissue. The clinical decision is made to have a cold knife cone excisional procedure (Slide 4).
- Take particular note of:
  - Nuclear atypia
  - Disruptions in the basement membrane
  - Relationship between atypical epithelium and underlying stroma


Benign ectocervix
HSIL
Disruptions in basement membrane
Invasive squamous cell carcinoma
Invasive squamous cell carcinoma

Patient Cervical Biopsy #3

	Biopsy #3	Biopsy #2	Biopsy #1	Normal
N:C ratio	High	High	Low	Low
Basal maturation defect	+++	+++	+	-
Nuclear atypia	High-grade	High-grade	“Koilocytic”	None
Disruption in the basement membrane	Broken	-	-	-
Relationship between atypical epithelium and underlying stroma	Invasion	None	None	None

Diagnosis?

- Squamous cell carcinoma

Learning Objectives

- **Describe the role of HPV in the pathogenesis of gynecologic tumors.**
- **Describe and identify the histopathologic features of HPV-related cervical dysplasia.**
- Explain and describe the relationship between inflammatory processes of the vulva and malignancy.
- Describe and identify infectious/inflammatory processes of the vulva and cervix.

Case 2

Case 2

- A 27-year-old patient presents to their gynecologist with a chief complaint of a two-week history of vaginal itching. They have had two weeks of vaginal itching with thick white clumpy discharge. They have had no other complaints, although they do note having had an upper respiratory infection treated with antibiotics a month ago. Prior to this episode, their gynecologic visits were unremarkable. Their gynecologist does a wet prep (see cervical image on next slide), checks the pH, and happens to take a cervical biopsy.

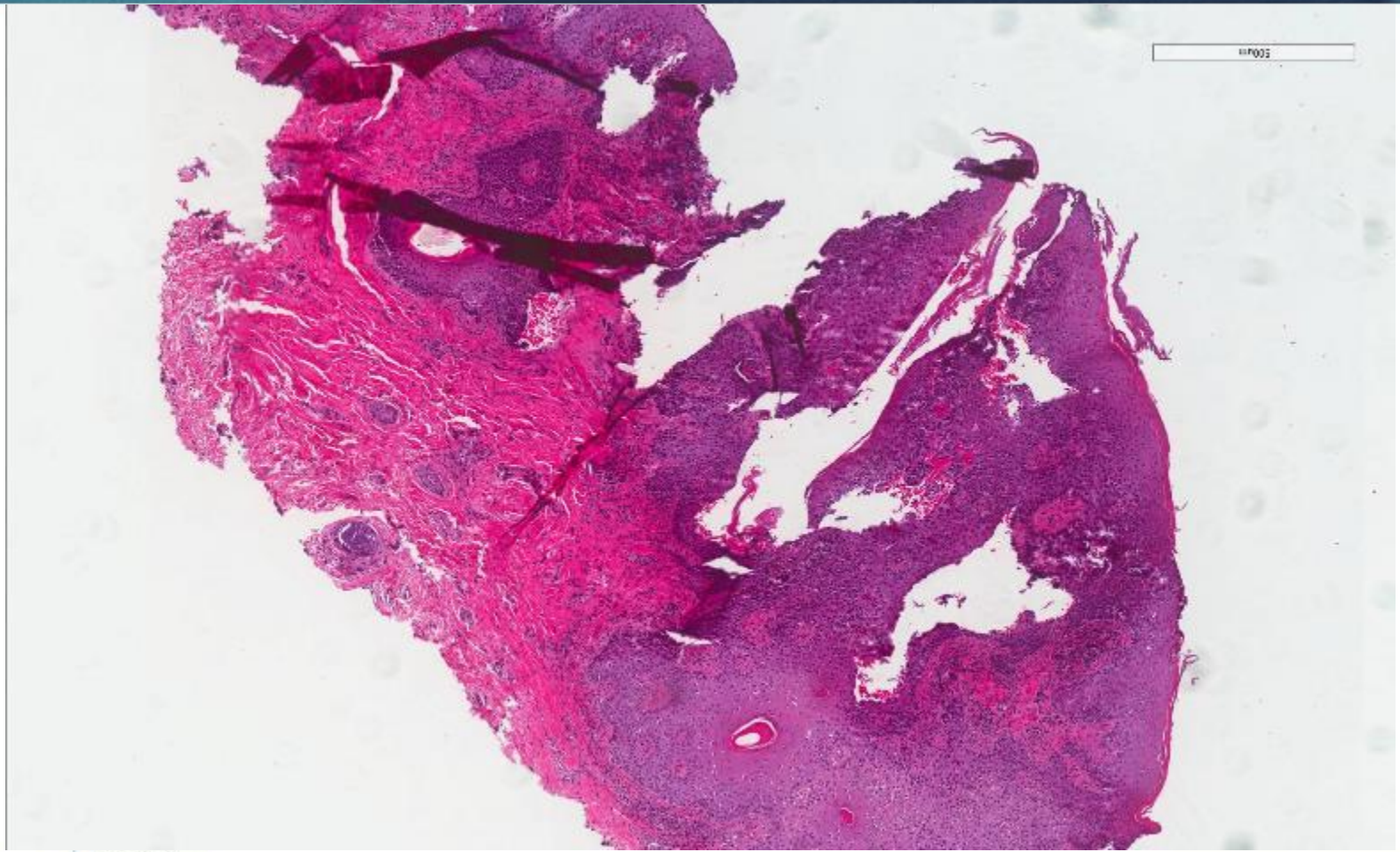
Case 2: Colposcopy

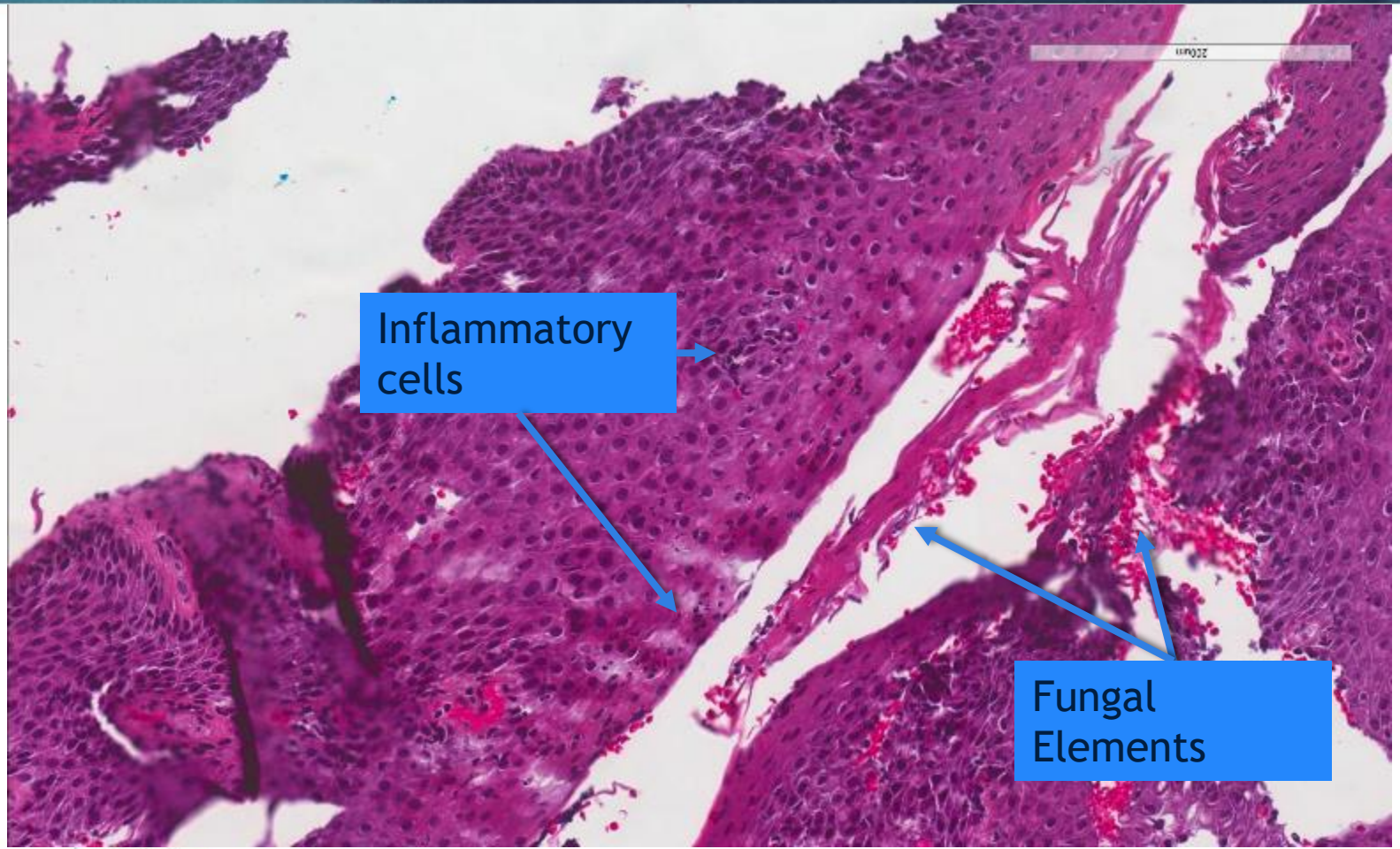


Histology time!



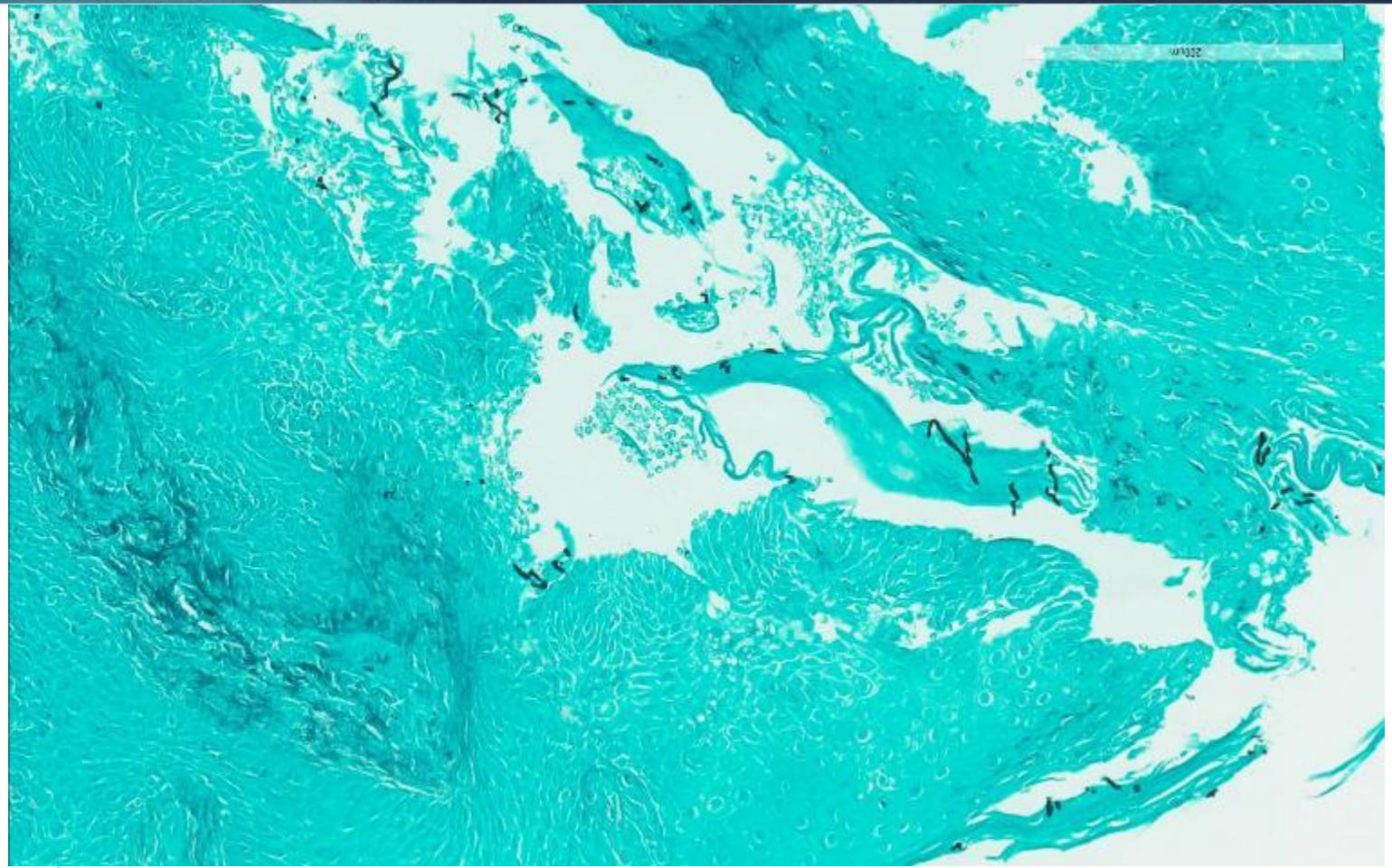
- Review the biopsy taken from the abnormal areas of the cervix (slide 5), taking special note of:
  - Presence or lack thereof of dysplasia
  - Presence of acute inflammatory cells (neutrophils)
  - Presence of fungal elements
- After reviewing slide 5, take a look at slide 6, which is a special stain for silver that makes fungal organisms look black under the microscope. Do you see fungus?




Inflammatory cells
Fungal Elements

Patient Cervical Biopsy vs. Normal Cervix

	Patient	Normal
Dysplasia	None	None
Neutrophils	Present	None
Fungal elements	Present	None



Histology time!



- Review the biopsy taken from the abnormal areas of the cervix (slide 5), taking special note of:
  - Presence or lack thereof of dysplasia –No dysplasia
  - Presence of acute inflammatory cells (neutrophils) –Loads of neutrophils
  - Presence of fungal elements –Technically yes, but a lot easier to see on the special stain (slide 6)
- After reviewing slide 5, take a look at slide 6, which is a special stain for silver that makes fungal organisms look black under the microscope. Do you see fungus?

Diagnosis?

- Fungal infection
  - Most commonly *Candida*
  - Risk factors
    - Immunosuppression
    - Prior antibiotic use
    - Douching
    - Increased estrogen levels
    - Uncontrolled diabetes

Learning Objectives

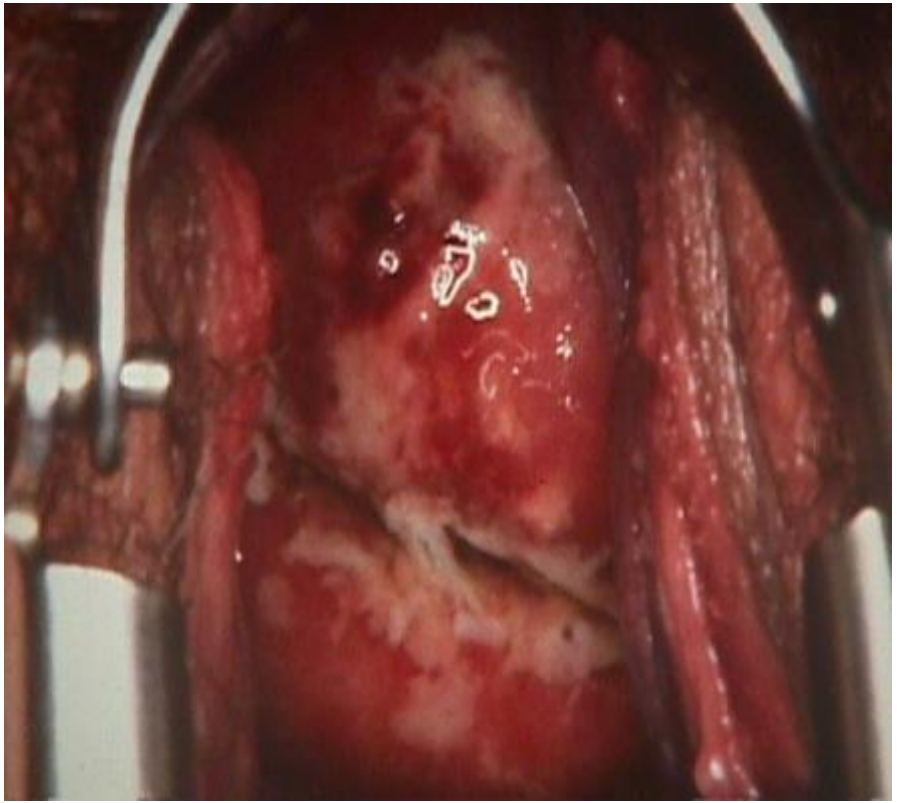
- Describe the role of HPV in the pathogenesis of gynecologic tumors.
- Describe and identify the histopathologic features of HPV-related cervical dysplasia.
- Explain and describe the relationship between inflammatory processes of the vulva and malignancy.
- **Describe and identify infectious/inflammatory processes of the vulva and cervix.**

Case 3

Case 3

- A 32-year-old patient presents to their gynecologist with a “painful rash” of the vulva for the last three days. They describe a three-day history of a diffuse burning, itching sensation. They also complain of a headache and general sense of feeling “unwell.”

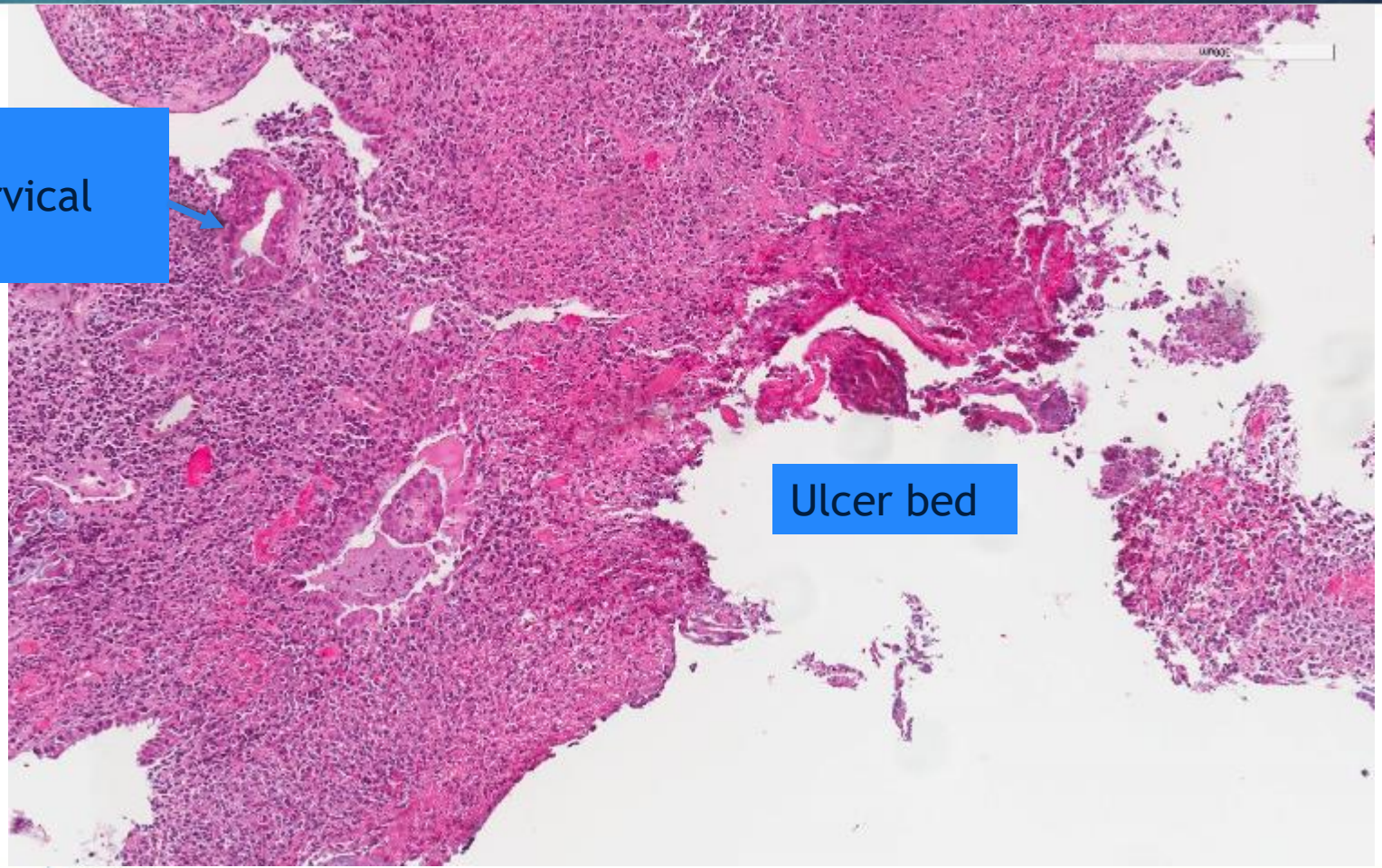
Case 3: Colposcopy



Histology time!



- Review the biopsy taken from the abnormal areas of the cervix (slide 7), taking special note of:
  - Presence of inflammatory cells
  - Presence of nuclear atypia
  - Presence of “normal” epithelium
- After reviewing slide 7, take a look at slide 8, which is a immunohistochemical stain for HSV-1 and -2, a stain that makes cells infected with HSV-1 or -2 turn dark brown. Do you see any evidence of HSV infection?


Benign endocervical glands
Ulcer bed

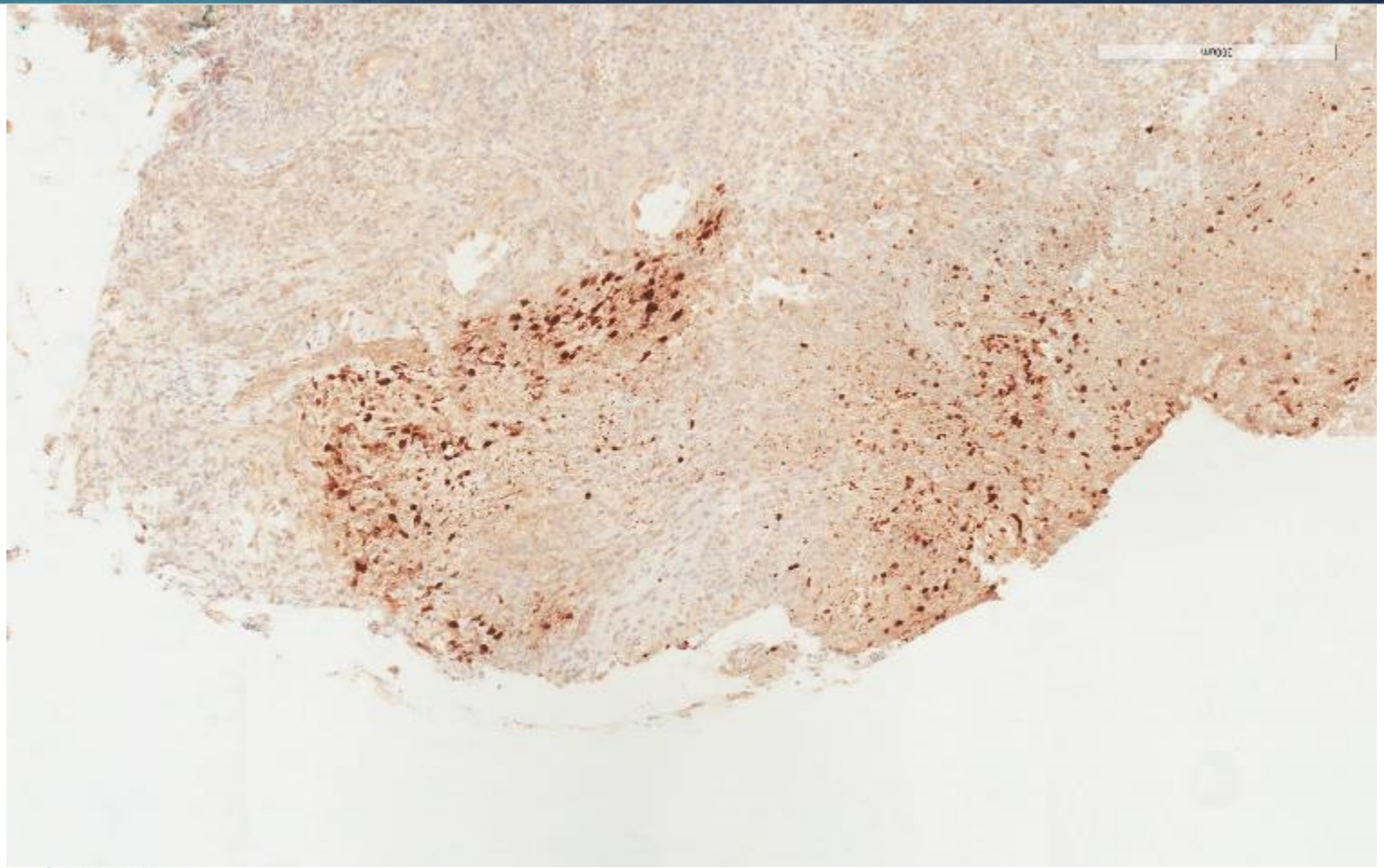
Patient Cervical Biopsy vs. Normal Cervix

	Patient	Normal
Nuclear atypia	None	None
Neutrophils	Present	None
Normal epithelium	Disrupted	Present

Histology time!

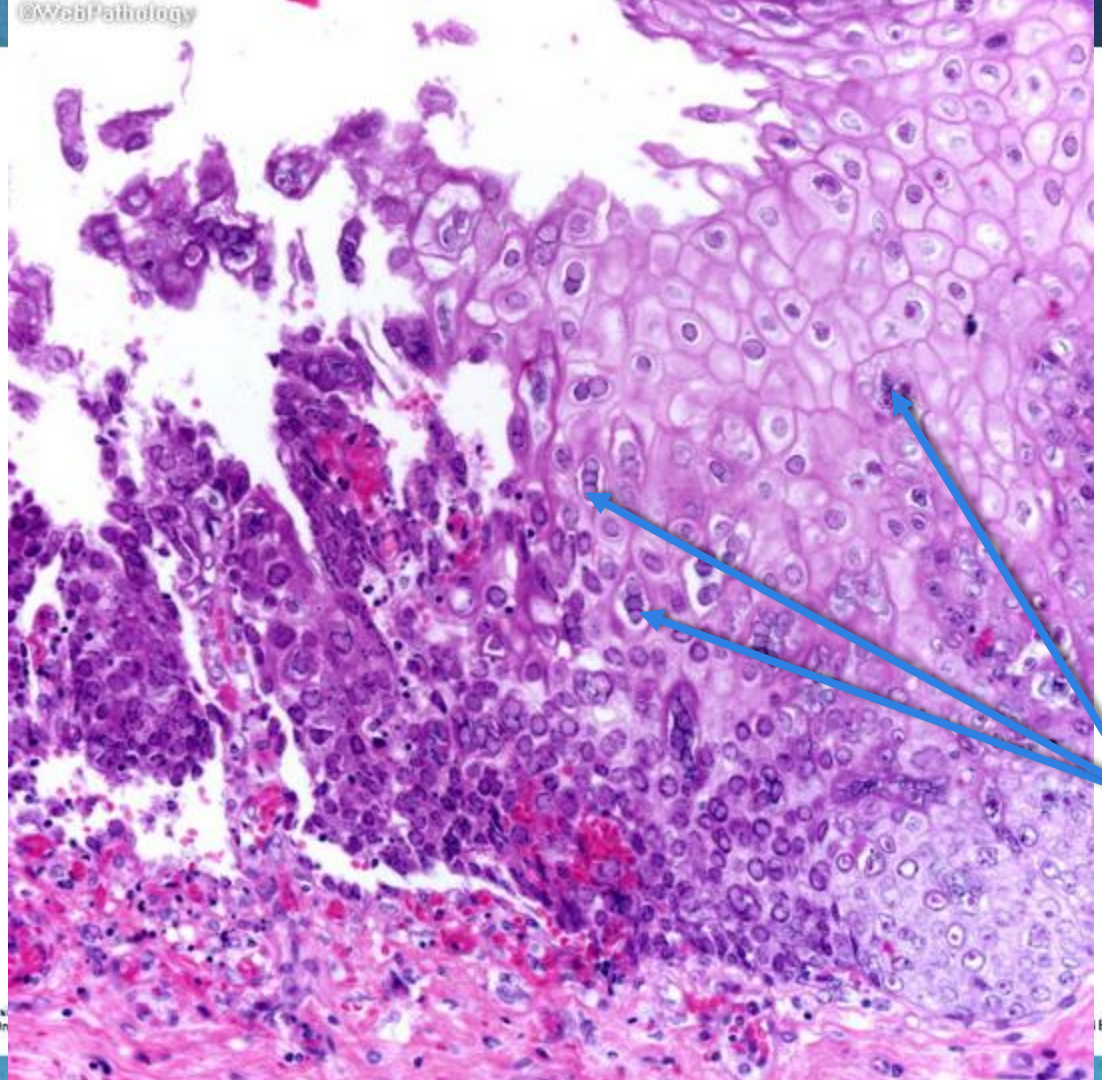


- Review the biopsy taken from the abnormal areas of the cervix (slide 7), taking special note of:
  - Presence of inflammatory cells – loads and loads, also associated with necrosis
  - Presence of nuclear atypia – Nope!
  - Presence of “normal” epithelium – Not on the surface
- After reviewing slide 7, take a look at slide 8, which is a immunohistochemical stain for HSV-1 and -2, a stain that makes cells infected with HSV-1 or -2 turn dark brown. Do you see any evidence of HSV infection?



Diagnosis?

- Herpes infection
  - Risk factors
    - Sex with person who has herpes infection
    - Improper use of barrier method
    - Prior sexually transmitted infections
    - Immunosuppression



Squamous Cells with Herpes viral inclusions (Cowdry bodies)

Learning Objectives

- Describe the role of HPV in the pathogenesis of gynecologic tumors.
- Describe and identify the histopathologic features of HPV-related cervical dysplasia.
- Explain and describe the relationship between inflammatory processes of the vulva and malignancy.
- **Describe and identify infectious/inflammatory processes of the vulva and cervix.**

Case 4

Case 4

- A 66-year-old patient presents to their gynecologist with a chief complaint of vulvar itching. They are not sure how long they have had the itching, but they think it got worse after menopause. On physical exam (see image below), their vulva is thin and fragile appearing, and the gynecologist takes a biopsy.

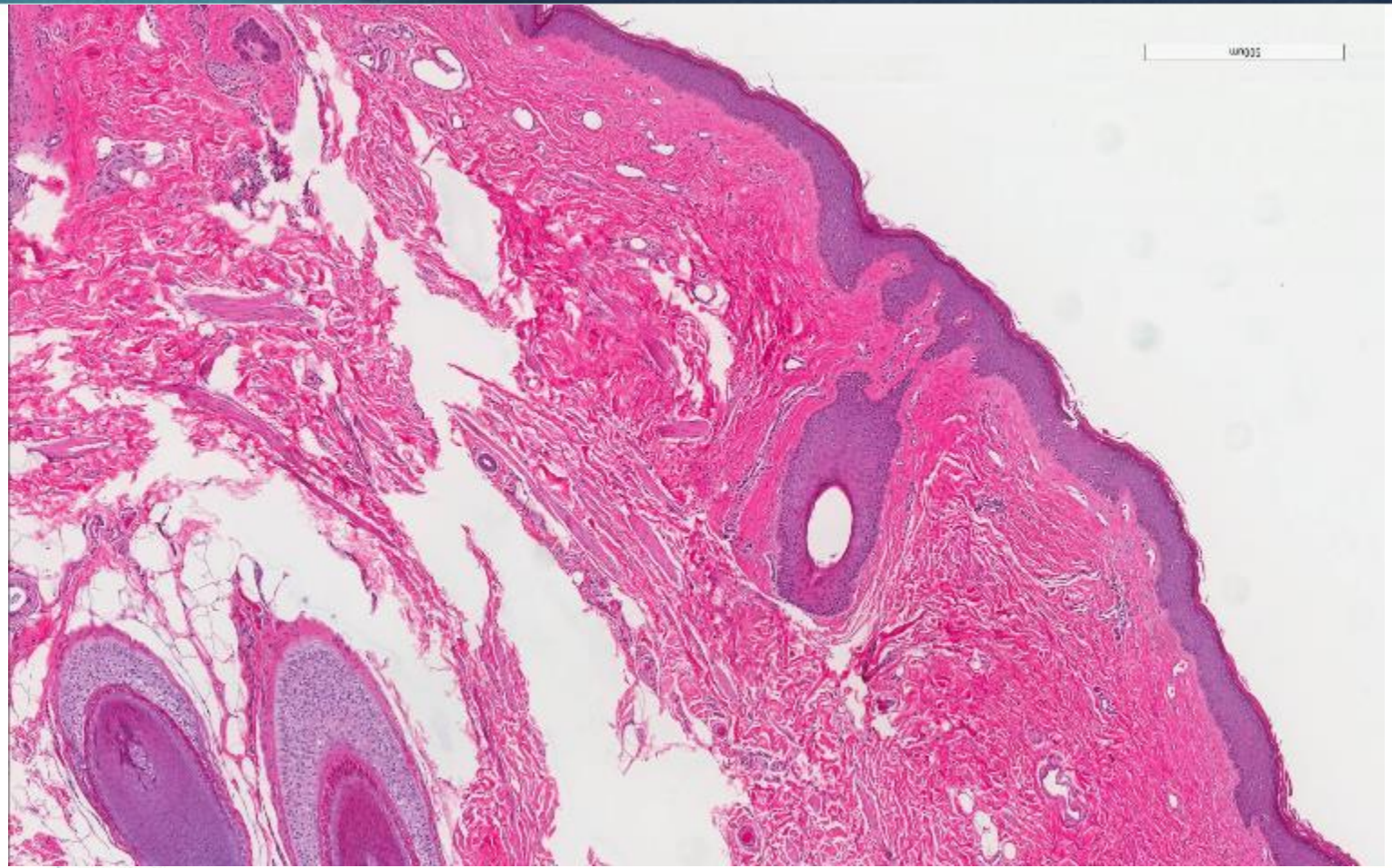


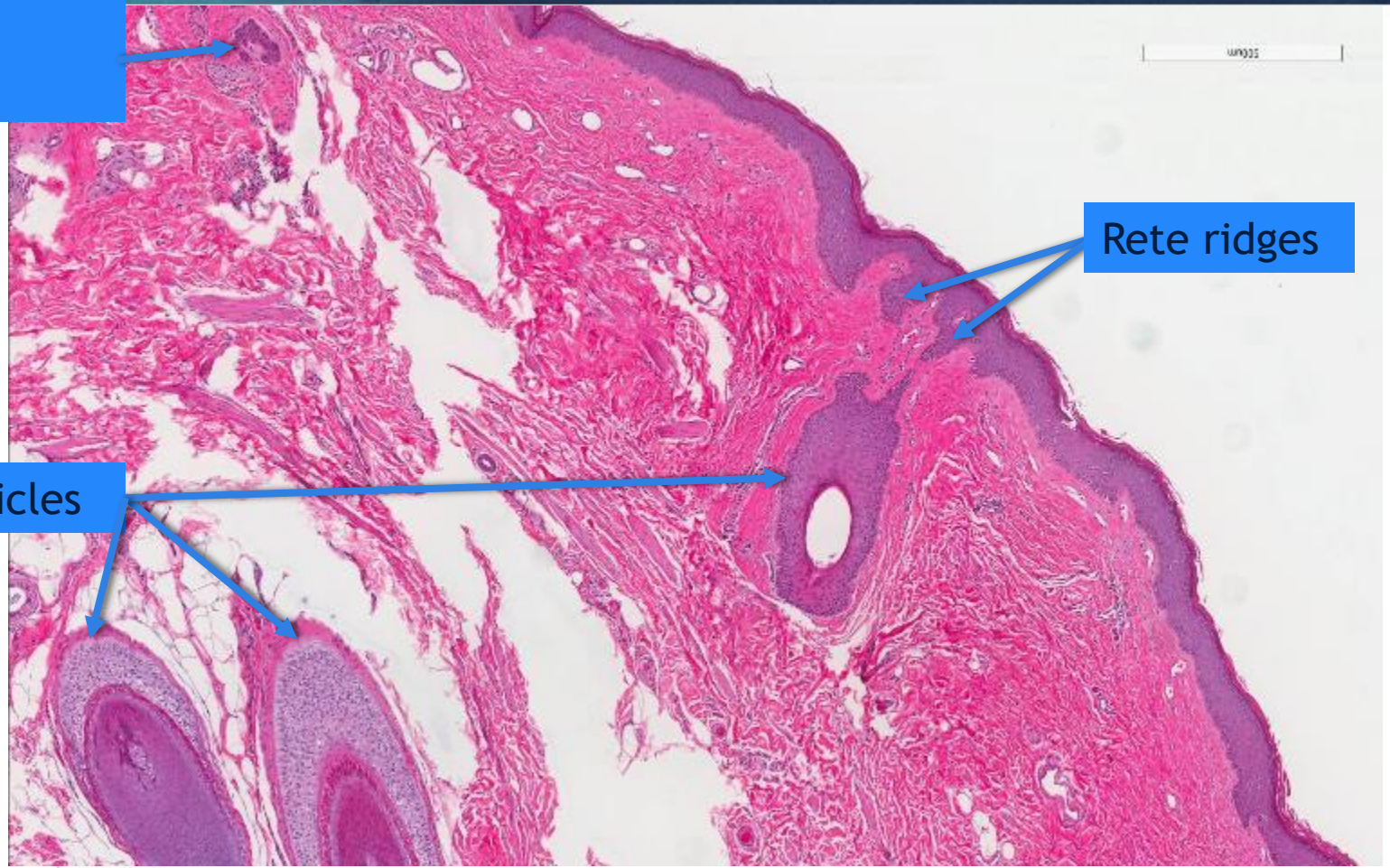


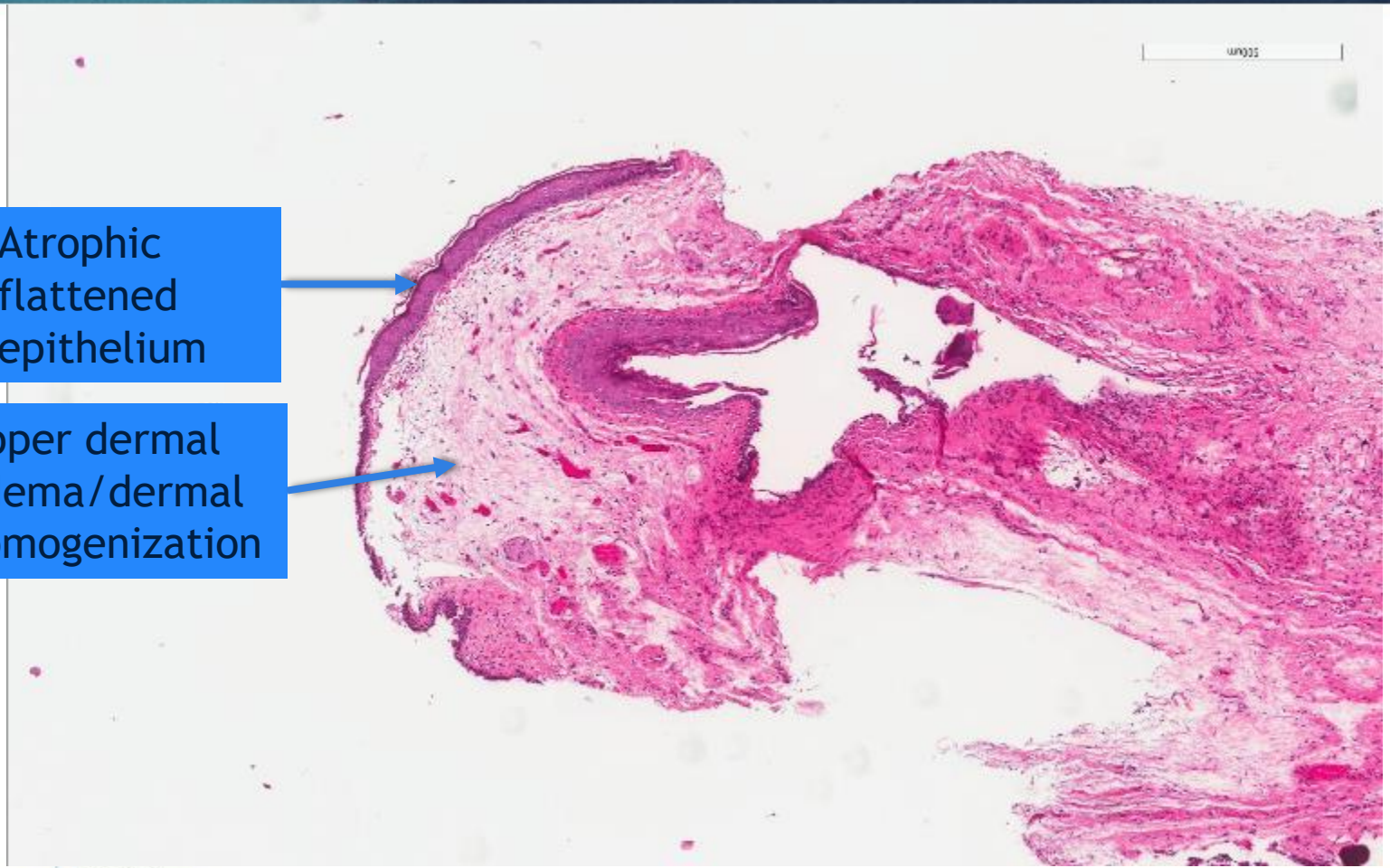
Histology time!



- Review the biopsy taken from the abnormal areas of the vulva (slide 10) and compare to normal vulva (slide 9), taking special note of:
  - Thinning of the epithelial layer
  - Lack of rete ridges
  - Homogenization of the dermal layer




Eccrine Glands
Rete ridges
Hair follicles


Atrophic flattened epithelium
Upper dermal edema/dermal homogenization

Case 4 Vulvar Biopsy vs. Normal Vulva

	Patient	Normal
Thickness of epithelial layer	Very thin	Normal
Rete ridges	Absent	Presence
Homogenization of dermal layer	Present	Absent

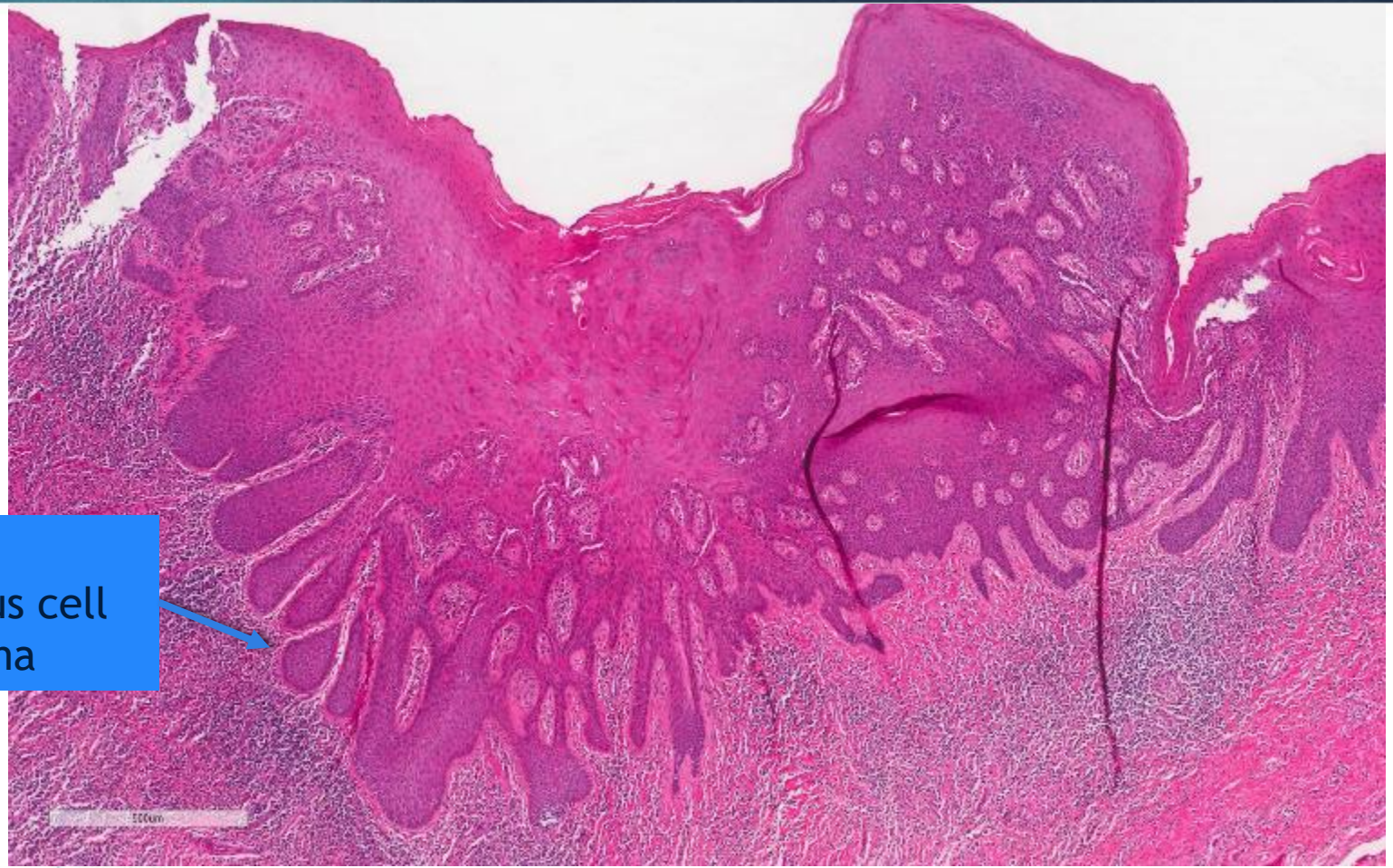
Diagnosis?

- Lichen Sclerosis (*et atrophicus)*
- LS is a risk factor for squamous cell carcinoma

Case 4: Continued (Part 2)



- On follow-up a year later, their vulva is still thin and fragile; however, there are nodular areas, which the gynecologist biopsies (slide 11). Note the following:
  - Similarity to prior biopsy specimen
  - Area of thickened epithelium with nuclear atypia
  - Interface between epithelium and underlying stroma


Invasive squamous cell carcinoma

Case 4 Vulvar Biopsy#2

	Vulva biopsy #2	Vulva Biopsy #1	Normal
Thickness of epithelial layer	Thickened	Very thin	Normal
Nuclear atypia	Present	Absent	Absent
Homogenization of dermal layer	Absent	Present	Absent
Relationship between atypical epithelium and underlying stroma	Invasive	None	None

Diagnosis?

- Squamous cell carcinoma
- *No association with HPV*

Learning Objectives

- Describe the role of HPV in the pathogenesis of gynecologic tumors.
- Describe and identify the histopathologic features of HPV-related cervical dysplasia.
- **Explain and describe the relationship between inflammatory processes of the vulva and malignancy.**
- Describe and identify infectious/inflammatory processes of the vulva and cervix.

Any Questions?

- Joanna.Chan@Jefferson.edu


Jefferson
Philadelphia University +
Thomas Jefferson University
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